IOT BASED GREEN HOUSE MONITORING AND CONTROL SYSTEM

AIM:

The main aim of this project is to implement IOT based greenhouse monitoring and control using Arduino.

PURPOSE:

General Agro related plants are gown outside environment. Sometimes fluctuations in environment effects on plant growth. We can't control plant growth when it was in open outside environment. But here we have solution like IOT based green house monitoring and controlling system.

DESCRIPTION:

This project consists WIFI (esp8266/IOT module) module through UART port. DHT11 (temperature and humidity) sensor, soil moisture sensor and LDR sensors are connected to Arduino through digital IO pins. Light, Fan and pump

WORKING:

Humidity, temperature, Soil moisture and LDR sensors are used for judging different parameters and information is communicated with microcontroller. The status of the sensors is displayed on the LCD. Also this information transmits to IOT server through WIFI (ESP8266 IOT) IOT module. Based on values user can control corresponding loads like fan, bulb, pump respectively from WIFI (Esp8266 IOT) IOT module. User can monitoring and control from anywhere.

TECHNICAL SPECIFICATIONS:

HARDWARE:

Microcontroller	:	Arduino Uno
Crystal	:	16 MHz
LCD	:	16X2 LCD
WIFI	:	Esp8266 (IOT module)
Temp/Humidity Sensor	:	DHT11
Relays	:	12V Electromagnetic coil
Pump	:	AC pump 230v
Fan	:	DC pump 12V
Light	:	230v AC
Power Source	:	12v 2 amp Adaptor

SOFTWARE:

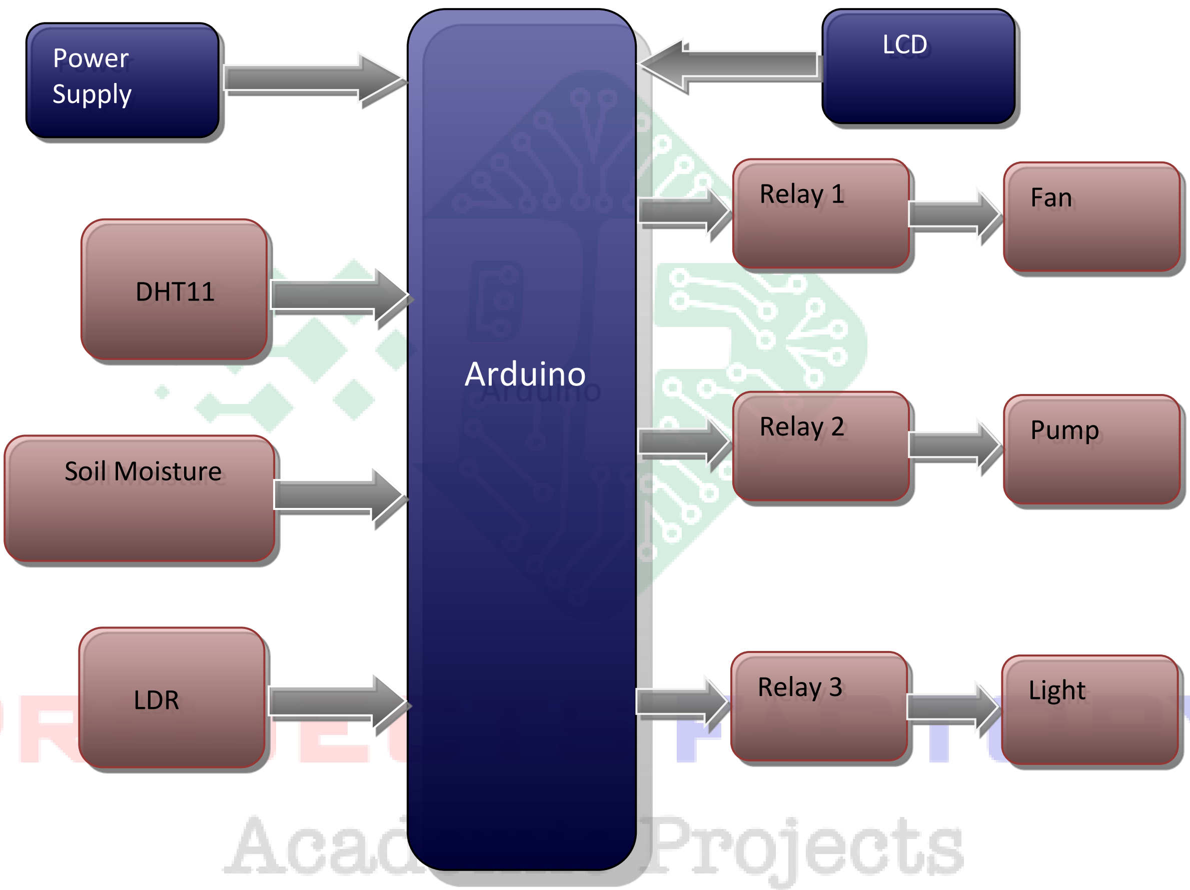
Arduino IDE

Proteus based circuit diagram

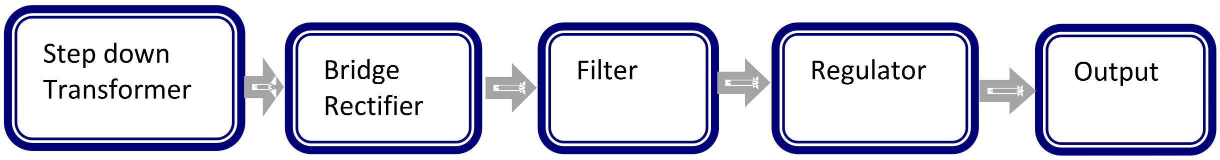
APPLICATIONS:

- For small scale agriculturists.

BLOCK DIAGRAM:



POWER SUPPLY BLOCKDIAGRAM:



INTERFACES COVERD:

- In this project we have covered WIFI (ESP8266/IOT) module interfacings. Also DHT11 (Temperature and humidity), Soil Moisture sensors and LDR along with relay Loads control.